20. Smithing, pottery and carpentry — making and repairing tools, vessels and structures

Oracle OS guide. Manuals in the library: "Forge Work" (Ilgen), "Practical Forging and Art Smithing", "The Potter's Craft", "Pottery for Artists", "How to Make Pottery", "Woodwork Joints", "Carpentry and Woodwork", "Rustic Carpentry", "Things To Make"; Gutenberg technology (illustrated, in Encyclopedias); Open Source Ecology.

Smithing (start with scrap)

- **Raw material**: car leaf springs, old files, rebar, rails, plate — there is salvageable metal for decades.
- **Forge**: **charcoal** fire in a pit or a stone/brick box with air blown from below (hide bellows, old hairdryer, crank blower). Anvil: heavy steel block, rail section or flat hard stone. Tongs, 1–1.5 kg hammer, bucket of water.
- **Steel colours** (in shade): dull red ~700 °C, orange ~900 °C (working), yellow ~1,100 °C (forge welding), white with sparks = burning (ruined).
- **Operations**: drawing out (lengthening), upsetting (shortening and thickening), bending, hot punching, cutting with a chisel, forge welding (borax or fine sand as flux).
- **Hardening** (carbon steel, e.g. springs, files): heat until a magnet no longer sticks, quench in oil/water → hard and brittle; then **temper**: clean and heat slowly to straw colour (knives, chisels) or blue (springs), cool.
- First useful items: nails, hooks, hinges, knives, chisels, hoe blades, horseshoes, forge tools.

Pottery

- **Clay**: sticky soil from riverbanks/ditch bottoms; dry, break up, soak, strain through cloth, drain to plasticine consistency; mix sand/ground fired clay (grog) to stop cracking.
- **Forming**: coils stacked and smoothed; slabs; moulding in a basket; kick wheel if possible.
- **Dry** slowly in shade 1–2 weeks (cracks = dried too fast).
- **Fire**: pit fire (600–800 °C, porous ware) or brick/clay kiln (900–1,100 °C, stronger); heat slowly over hours.
- **Waterproofing**: burnish before firing, glaze (ash + clay + sand in a hot kiln), or seal with milk/fat.
- **Bricks**: clay + straw/sand in wooden moulds, sun-dried (adobe) or clamp-fired (brick).

Hand-tool carpentry

- **Basic tools**: axe, crosscut and rip saws, chisels, plane, brace and bit, mallet, square, plumb line, water level (clear hose with water), file, sharpening stone.
- **Sharpening**: 25–30° angle, wet stone, steady strokes until a burr forms; remove it on the other side.

- **Timber**: fell in winter, season 1 year per 2.5 cm thickness, stacked with spacers, in shade.
- **Joints** (no nails): half-lap, mortise and tenon (strongest for frames), dovetail (drawers and boxes), wooden pegs.
- **Priority items**: tool handles (ash), gates, benches and tables, beds, window frames, beehives, wheelbarrows, wheels, barrels.
- **Green wood**: carve spoons and bowls, peg-jointed chairs (easy to work fresh).

Safety

Smithing: goggles, leather apron, closed shoes, water nearby, never quench indoors. Pottery/kilns: fire outdoors, watch smoke and burns. Carpentry: always cut away from the body; sharp tools are safer than blunt ones.